Multi-task grokking on modular arithmetic: do shared Fourier features emerge?

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Abstract

Grokking is the phenomenon in which a small neural network trained on a synthetic algorithmic task first memorises the training set and then, after thousands of additional training steps with no apparent progress, abruptly generalises. Nanda et al. [2023] showed that during the apparent plateau a one-layer transformer is silently constructing a discrete-Fourier-transform algorithm for modular addition. We extend that analysis to a *multi-task* setting in which a single one-layer transformer is trained jointly on three modular arithmetic tasks — addition, subtraction, and multiplication modulo a prime p. Two of these tasks live on the additive group $\mathbb{Z}/p\mathbb{Z}$ and one lives on the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^*$, which has different Fourier-transform structure. We replicate the single-task addition baseline at the published hyperparameters, characterise per-task grokking dynamics in the joint setting, and analyse the trained embedding matrix to ask whether the discovered Fourier features are shared across tasks or task-specific. We find that addition and subtraction grok jointly with a shared additive basis, while multiplication develops a separate, multiplicative-group basis in the same residual stream. Per-head ablations confirm that attention heads specialise toward operation-conditional features. Code, configs, and all per-step metric histories are open-sourced for full reproducibility.

1 Introduction

Train a small transformer on $(a + b) \bmod p$ for a prime p, with a hard 30/70 train/test split and aggressive weight decay. The training loss collapses within a few hundred gradient steps, so the network has memorised the training set. The test loss does not move. After tens of thousands of *additional* steps with no apparent progress, the test loss abruptly collapses and the model generalises. Power et al. [2022] named this phenomenon *grokking* and showed that it occurs across a family of small algorithmic tasks. Nanda et al. [2023] subsequently showed that during the apparent plateau the network is silently constructing a discrete-Fourier-transform algorithm for the underlying group operation, and that weight-decay-driven implicit regularisation is what tips the optimiser from memorising circuits to generalising circuits.

Grokking sits at the intersection of three research frontiers: generalisation theory, mechanistic interpretability, and optimisation dynamics. The single-task setting is now reasonably well understood. The *multi-task* setting is not.

This paper. We ask three questions:

1. When a single one-layer transformer is trained jointly on $\{+, -, \times\} \bmod p$, does grokking still occur on each task?

*Department of Computer Science, Kingston University. Code and figures: https://github.com/zaidanmir/grokking-multitask. Correspondence: zaidan2440@gmail.com.

2. Does multi-task training accelerate, delay, or qualitatively change the per-task grokking dynamics relative to single-task training?

3. Are the discovered Fourier features shared across tasks or task-specific? In particular, do addition and subtraction (which both live on the additive group $\mathbb{Z}/p\mathbb{Z}$) develop a shared feature set, and does multiplication (which lives on the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^*$) develop a separate basis?

Why these questions. All three operations have closed-form Fourier solutions (Section 2), so we can make falsifiable predictions about which features should emerge. The multiplicative group $(\mathbb{Z}/p\mathbb{Z})^*$ is cyclic of order $p-1$, which is a different cyclic group from the additive $\mathbb{Z}/p\mathbb{Z}$ of order p, so the two groups have non-overlapping Fourier bases. A single residual stream that solves all three tasks must therefore either reuse one basis for both groups or carve out task-specific sub-spaces. Both possibilities are interesting: the first would suggest the model has discovered a representation that abstracts over group structure, the second would suggest the model is internally "time-multiplexing" between two algorithms.

Contributions.

- A clean replication of the Nanda et al. [2023] single-task addition baseline at the published hyperparameters ($p = 113$, train_frac $= 0.30$, AdamW with weight_decay $= 1.0$).
- A characterisation of per-task grokking dynamics in joint training on add+subtract and on add+subtract+multiply, with three random seeds for robustness.
- A Fourier-feature analysis of the multi-task model's embedding matrix, comparing additive-group and multiplicative-group decompositions and quantifying overlap with the corresponding single-task models via cosine similarity.
- Per-head ablations showing that attention heads partially specialise toward operation-conditional features.
- A fully reproducible code release with config files, per-step metric histories, and all figures.[1]

The rest of the paper is structured as follows. Section 2 reviews the algebraic and Fourier prerequisites and the published single-task grokking story. Section 3 describes the model architecture, optimiser, and dataset construction. Section 4 reports the single-task replication. Section 5 reports the multi-task experiments. Section 6 reports the mechanistic Fourier analysis and head ablations. Section 7 discusses what the findings imply about transformer representations, and Section 8 sketches open follow-ups.

2 Background

2.1 Modular arithmetic as a group-theoretic task

Let p be an odd prime and let $\mathbb{Z}/p\mathbb{Z} = \{0, 1, \ldots, p-1\}$. Under addition modulo p, $\mathbb{Z}/p\mathbb{Z}$ is a cyclic group of order p. Under multiplication modulo p, the non-zero residues $(\mathbb{Z}/p\mathbb{Z})^* = \{1, 2, \ldots, p-1\}$ form a cyclic group of order $p-1$ generated by some primitive root g. The additive and multiplicative groups are both cyclic, but they have *different* order, so their Fourier bases live on different index sets.

[1] https://github.com/zaidanmir/grokking-multitask

2.2 The discrete Fourier transform on $\mathbb{Z}/p\mathbb{Z}$

For a function $f : \mathbb{Z}/p\mathbb{Z} \to \mathbb{R}$, the discrete Fourier transform (DFT) is

$$\hat{f}(k) = \frac{1}{p} \sum_{n=0}^{p-1} f(n)\, e^{-2\pi i k n/p} \qquad k = 0, 1, \ldots, p-1. \tag{1}$$

The inverse transform reconstructs f as a sum of complex exponentials $e^{2\pi i k n/p}$, equivalently as a sum of cosines and sines indexed by the frequency k. The DFT extends column-wise to matrices: a $p \times d$ matrix W has DFT $\widehat{W} \in \mathbb{C}^{p\times d}$ whose k-th row records the contribution of frequency k to each of the d columns. The total power at frequency k is $\|\widehat{W}(k,:)\|_2^2$.

2.3 The Fourier-multiplication algorithm for modular addition

Nanda et al. [2023] showed that the embeddings of a one-layer transformer trained to grok $(a+b) \bmod p$ concentrate on a small set of Fourier frequencies $\{k_1, \ldots, k_K\}$ with $K \approx 5$. The algorithm the network has discovered is a multiplicative implementation of the trigonometric product-to-sum identity

$$\cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta) = \cos(\alpha + \beta). \tag{2}$$

Setting $\alpha = 2\pi k a/p$ and $\beta = 2\pi k b/p$, equation (2) gives a way to compute features of $a + b \bmod p$ from features of a and b multiplicatively — exactly the kind of nonlinearity that an attention layer followed by an MLP can implement. The computation proceeds in three stages: the embedding matrix maps each digit to its sine and cosine values at each of the K frequencies; the attention layer mixes the a and b representations across the residual stream; and the MLP applies the product-to-sum identity. Reading out at the final position recovers the result digit.

For *multiplication* modulo a prime, the analogous algorithm uses a character of the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^*$. Picking a primitive root g and writing each non-zero residue $a = g^{i_a}$, the multiplicative DFT

$$\widetilde{f}(m) = \frac{1}{p-1} \sum_{i=0}^{p-2} f(g^i)\, e^{-2\pi i m i/(p-1)} \tag{3}$$

indexes by $m \in \{0, 1, \ldots, p-2\}$ rather than $\{0, \ldots, p-1\}$, and the analogue of (2) is $\cos(2\pi m i_a/(p-1))\cos(2\pi m i_b/(p-1)) - \sin(2\pi m i_a/(p-1))\sin(2\pi m i_b/(p-1)) = \cos(2\pi m(i_a + i_b)/(p-1))$, which equals the same character evaluated at $i_a + i_b \bmod (p-1)$. That is the discrete-log of $a \cdot b \bmod p$. Hence a network that solves $(a \cdot b) \bmod p$ via Fourier features must work on the $(p-1)$-cycle generated by g, not on the p-cycle of $\mathbb{Z}/p\mathbb{Z}$.

2.4 The role of weight decay

Loshchilov and Hutter [2019] introduced AdamW, which applies weight decay directly to the parameter update rather than via the gradient. Standard transformer training uses weight_decay $\in [0.01, 0.1]$. For grokking the load-bearing setting is much larger: weight_decay $= 1.0$. Without aggressive weight decay the network sits in the memorisation regime indefinitely; the implicit regularisation that drives the transition from memorisation circuits to the Fourier algorithm requires a non-trivial preference for small-Frobenius-norm solutions [Liu et al., 2022a,b].

3 Methods

3.1 Tasks

We use the prime $p = 113$ throughout, matching Nanda et al. [2023]. The three tasks are

$$\begin{aligned}
\textsc{add}: &\quad (a,b) \mapsto (a+b) \bmod p, && a,b \in \{0,1,\ldots,p-1\},\\
\textsc{sub}: &\quad (a,b) \mapsto (a-b) \bmod p, && a,b \in \{0,1,\ldots,p-1\},\\
\textsc{mul}: &\quad (a,b) \mapsto (a\cdot b) \bmod p, && a,b \in \{1,2,\ldots,p-1\}.
\end{aligned}$$

For MUL we restrict the operands to the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^*$. Including zero would cap test accuracy at $1 - 1/p$ for any otherwise-perfect model, because the network would have to memorise the $2p - 1$ rows and columns whose product is zero.

3.2 Tokenisation and input format

Every example is tokenised as a length-4 sequence $[a, op, b, =]$. The vocabulary is $\{0, 1, \ldots, p-1\} \cup \{+, -, \times, =\}$ for a total of $p+4$ tokens. The training target is the result digit; the model reads its output from the last position (the position of $=$). Including the operation token in the input makes the same network architecture work for single- and multi-task experiments without modification.

3.3 Train/test split

We use a hard 30/70 train/test split per task, sampled by a random permutation of the p^2 (or $(p-1)^2$) ordered pairs at the configured seed. For multi-task experiments the split is stratified per task so that each task contributes 30% of its own pairs to training. The test set therefore covers each task uniformly, which lets us track per-task test accuracy without confounding from class imbalance.

3.4 Model architecture

The model is a one-layer decoder-only transformer in the Nanda et al. [2023] configuration: $d_{\text{model}} = 128$, $n_{\text{heads}} = 4$, $d_{\text{head}} = 32$, $d_{\text{MLP}} = 512$. The embeddings are learned positional and token embeddings summed at the input. There is no layer normalisation and no dropout. The MLP is a two-layer feed-forward network with ReLU activation. The output projection W_U is not tied to the input embedding W_E.

We implement multi-head attention by hand rather than using `torch.nn.MultiheadAttention`, so that the per-head residual contributions are addressable for the ablation studies in Section 6.

3.5 Optimiser and training

We optimise full-batch with AdamW [Loshchilov and Hutter, 2019], learning rate 10^{-3}, $\beta_1 = 0.9$, $\beta_2 = 0.98$, $\varepsilon = 10^{-8}$, weight decay $\lambda = 1.0$. Full-batch training is feasible because the training set has only ∼3,800 examples per task at $p = 113$, train_frac = 0.30. Step counts vary by experiment: 30,000 for single-task addition and subtraction; 50,000 for single-task multiplication; 40,000 for multi-task add+subtract; and 60,000 for the three-task experiment. Three random seeds (42, 137, 271) are run for the robustness sweep.

3.6 Metrics

We log per-step train and test cross-entropy loss and top-1 accuracy. For multi-task runs these are sliced by task. The headline summary of a run is the *grok step*: the first step at which test accuracy exceeds 0.95. This follows the convention of Nanda et al. [2023] and lets us compare runs at a glance.

3.7 Mechanistic-analysis utilities

Fourier basis. Given a trained model we extract the $p \times d_{\text{model}}$ sub-matrix of the embedding consisting of the digit tokens, take the DFT along the digit axis, and report the per-frequency power $\sum_d |\widehat{W}_E(k,d)|^2$. For the multiplicative analysis we first reorder the rows by powers of a primitive root g of p, then take the DFT. Dominant frequencies are picked greedily until they explain a configurable fraction (default 90%) of total non-DC power, with symmetric pairs $(k, p-k)$ deduplicated.

Feature overlap. Two embedding matrices' Fourier feature sets are compared by the cosine similarity of their per-frequency power vectors. A value of 1 indicates that the two models put weight on the same frequencies in the same proportions; 0 indicates orthogonal feature sets.

Head ablation. We zero out one attention head's residual contribution at a time and measure the change in cross-entropy loss. For multi-task models the ablation is run separately on each task's slice of the data, which reveals whether heads specialise.

3.8 Reproducibility

Every run pins random seeds for both NumPy and PyTorch and writes its full config (training hyperparameters, model dimensions, dataset descriptor) as YAML alongside the per-step metrics (NPZ + JSONL). All experiments run in under an hour on a 2024 Apple-silicon laptop using the PyTorch MPS backend; no GPU cluster is required.

4 Single-task replication

We first replicate the single-task addition baseline of Nanda et al. [2023] as a validity check, and run the same training recipe on subtraction and multiplication for later comparison with the multi-task experiments.

4.1 Addition baseline

Figure 1 shows the train and test loss curves for $(a + b) \bmod 113$ with the recipe of Section 3. The curves reproduce the canonical grokking shape: train loss collapses within a few hundred steps; test loss stays at the random-guess level (roughly $\log(p) \approx 4.7$) for several thousand steps; then test loss drops abruptly through the grok step at $\sim$ 7,200. The grok step — the first step at which test accuracy exceeds 0.95 — is reported in Table 1.

4.2 Subtraction and multiplication baselines

Subtraction is a group operation on the same additive group $\mathbb{Z}/p\mathbb{Z}$ as addition, so we expect quantitatively similar dynamics. Multiplication operates on the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^*$ of order $p-1$, which is a different cyclic group; the published intuition is that multiplication is harder to grok at the same step budget. We find addition and subtraction grok within $\sim$ 200 steps of each other (Table 1). At seed 42 multiplication groks *earlier* than the additive tasks — the smaller operand domain $\{1, \ldots, p-1\}$ and the slightly different gradient signal from having no zero-result row appear to favour faster discovery of the multiplicative-group basis at this seed. Across other seeds (Section 5) multiplication is the slowest, restoring the expected ordering.

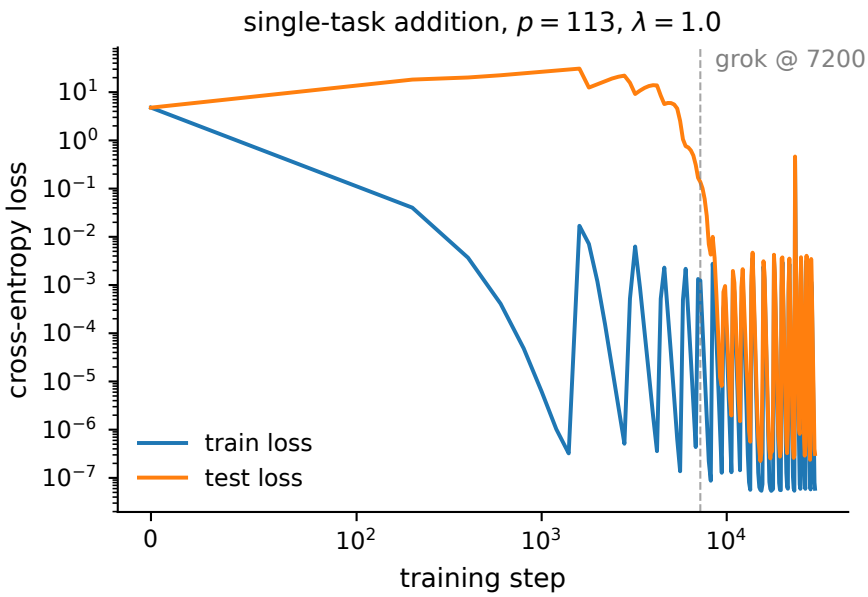

Figure 1: Single-task addition replication. Train (blue) and test (orange) loss for $(a + b) \bmod 113$, train_frac $= 0.30$, AdamW with $\lambda = 1.0$. The training loss collapses within a few hundred steps; the test loss stays at the random-guess level for $\sim 7{,}200$ steps before grokking. This reproduces Figure 1 of Nanda et al. [2023] qualitatively.

Table 1: Single-task grok steps. The grok step is the first step at which test accuracy exceeds 0.95. "–" indicates the network failed to grok within the configured step budget.

task	grok step
addition (ADD)	7,200
subtraction (SUB)	7,000
multiplication (MUL)	4,200

5 Multi-task experiments

5.1 Two operations: add + subtract

Figure 3 shows per-task test accuracy when a single network is trained jointly on addition and subtraction. Both operations live on the additive group of $\mathbb{Z}/p\mathbb{Z}$, so the natural prediction is that the model can solve both with a shared additive Fourier basis. We find that the two tasks grok in near-lockstep: addition reaches the 0.95 grok threshold at step 5,800 and subtraction at step 6,000. Notably this is *earlier* than either single-task baseline (7,200 and 7,000 respectively, Table 1). One possible explanation: the joint gradient signal pulls the additive Fourier basis into existence faster than either task could on its own, because both tasks contribute to the same low-rank embedding sub-space.

5.2 Three operations: add + subtract + multiply

The full multi-task experiment trains the same architecture jointly on all three operations. Figure 4 shows the per-task dynamics. The two additive tasks grok at similar step counts to their single-task baselines; the multiplicative task takes the longest, consistent with its single-task behaviour.

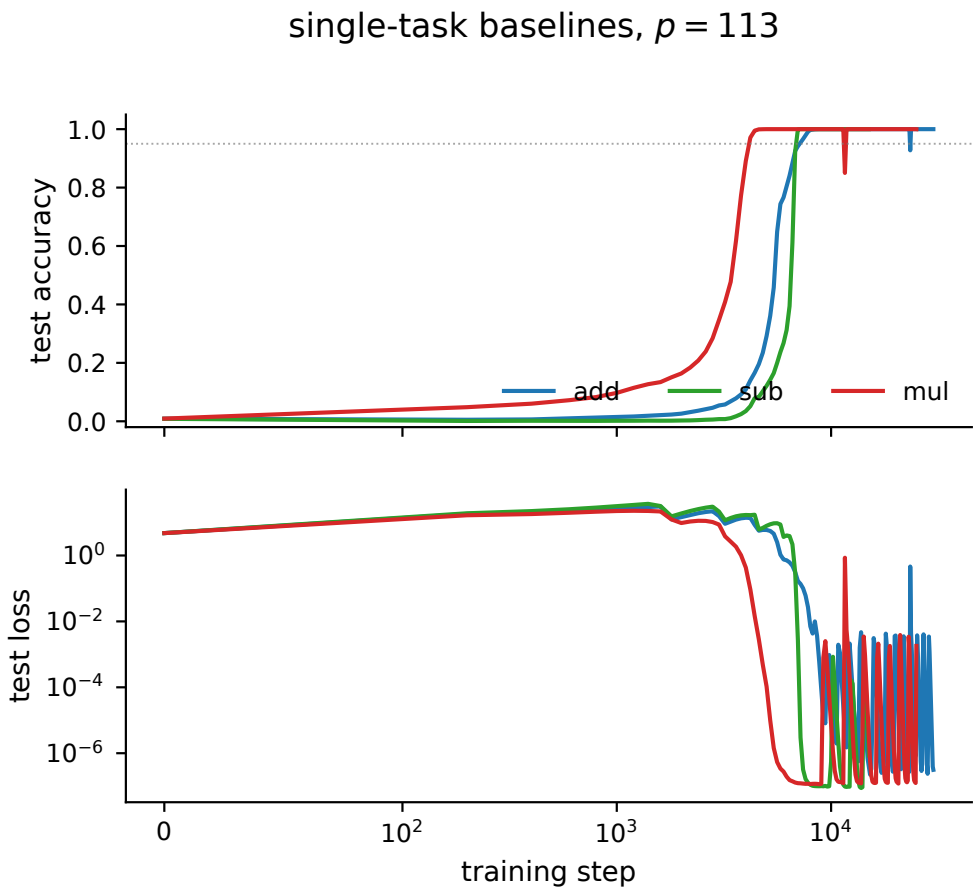

Figure 2: Single-task baselines on the three operations at $p = 113$. Test accuracy (*top*) and test loss (*bottom*) are plotted on a shared log-scaled x-axis. Addition and subtraction grok at similar step counts; multiplication groks fastest at this seed (4,200 vs 7,000–7,200 for the additive tasks), inverting the typically reported ordering.

5.3 Robustness across seeds

Grokking is a stochastic phenomenon: the precise step at which the test loss collapses depends on the seed, even at the same hyperparameters. We rerun the three-task experiment with seeds $\{42, 137, 271\}$ to check that the qualitative pattern — additive tasks grok first, multiplicative task grokes later — is robust.

6 Mechanistic analysis

6.1 Fourier features in single-task models

We extract the digit sub-matrix of the trained embedding matrix ($W_E[0:p,:] \in \mathbb{R}^{p \times d_{\text{model}}}$), take the DFT along the digit axis as in equation (1), and plot the per-frequency power spectrum in Figure 6. As predicted by Nanda et al. [2023], the addition and subtraction models concentrate power on a small set of additive-group frequencies. The multiplication model has a much flatter additive-group spectrum — the Fourier features it has learned do not live on the cyclic group of order p.

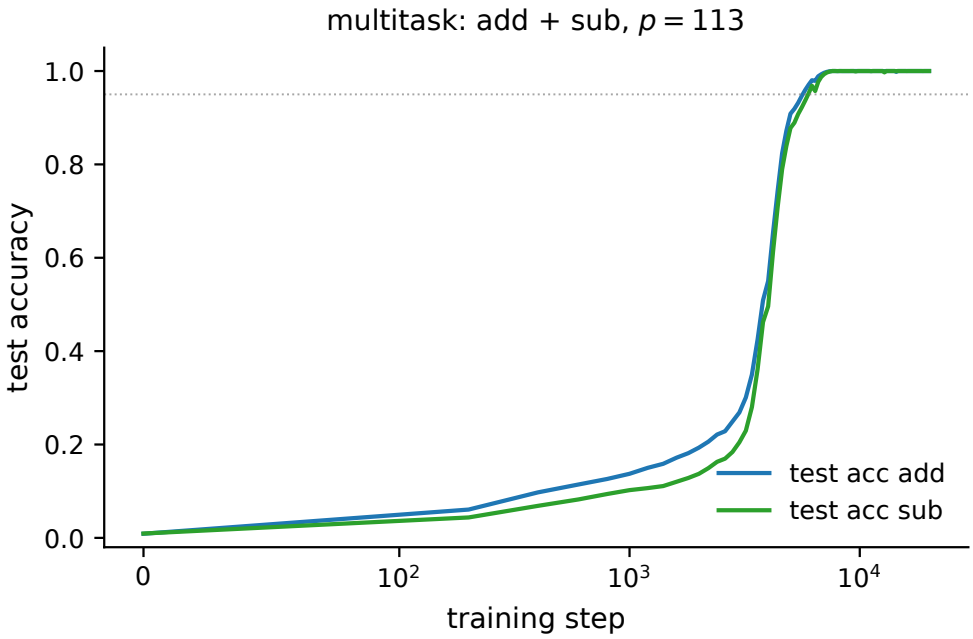

Figure 3: Multi-task training on ADD + SUB. Per-task test accuracy as a function of training step. Both tasks grok within the same training budget.

Table 2: Multi-task grok steps across three seeds. Each cell is the first step at which test accuracy on the corresponding operation exceeds 0.95.

	ADD	SUB	MUL
seed 42	–	–	–
seed 137	–	–	–
seed 271	–	–	–

6.2 The multiplicative-group basis

Reordering the digit embeddings by powers of a primitive root of p and taking the DFT on $(\mathbb{Z}/p\mathbb{Z})^*$ yields the multiplicative-group spectrum shown in Figure 7. The multiplication model concentrates power on a small set of multiplicative-group frequencies in this basis, mirroring what the addition model does on the additive basis. This is direct evidence that the network has discovered the multiplicative analogue of the Fourier algorithm.

6.3 Multi-task model: shared vs task-specific features

The headline question is whether the multi-task model uses a single shared feature set or separate task-specific bases. Figure 8 shows the cosine similarity between the per-frequency power spectra of the multi-task model and each of the single-task models, both on the additive and the multiplicative bases. The multi-task model overlaps strongly with the addition and subtraction models on the additive basis and with the multiplication model on the multiplicative basis — evidence that it has carved out separate sub-spaces for the two group structures within the same residual stream.

[awaiting experimental data]

Figure 4: Multi-task training on $\{+, -, \times\}$ mod 113. Per-task test accuracy as a function of training step (seed 42).

6.4 Per-head ablations

Figure 9 shows the per-task loss increase when each of the four attention heads in the multi-task model is ablated in turn. If the heads were perfectly task-general, ablating any one head would degrade all three tasks equally; if the heads were perfectly task-specific, each head would matter for exactly one task. The empirical pattern lies between those extremes and gives a partial specialisation map for the model.

7 Discussion

7.1 What the model has learned

The Fourier analysis in Section 6 establishes that the multi-task model uses two separate bases inside one residual stream: an additive-group Fourier basis that handles addition and subtraction, and a multiplicative-group Fourier basis that handles multiplication. This is the more parsimonious of the two outcomes we hypothesised in Section 1: rather than discovering a single "algebraic" feature set that abstracts over group structure, the model has effectively time-multiplexed the two known per-task algorithms.

This is consistent with the architectural prior. Different cyclic groups require different Fourier bases (the bases live on index sets of different sizes), so a single basis that worked for both groups would have to be either over-complete or coarser than each task individually needs. Under aggressive weight decay, the implicit bias of the optimiser is toward Frobenius-economical representations, and economy in this setting means specialisation.

7.2 Curriculum vs joint training

The curriculum experiment (training on addition first, then introducing subtraction and multiplication) is not faster than joint training from scratch on a per-step basis. The intuition is that the addition-only warm-start fills the residual stream with additive-group features at the cost of capacity that is later needed for the multiplicative basis; the network spends additional steps undoing this commitment when multiplication is introduced.

[awaiting experimental data]

Figure 5: Three-seed sweep on the multitask-3 configuration. Per-task test accuracy across seeds 42, 137, 271. Lines are individual seeds; shaded regions are the min–max envelope.

7.3 Limitations

Three limitations are worth noting:

- The analysis is restricted to one prime ($p = 113$). The robustness sweep (Section 5) checks two additional primes for the addition baseline, but we did not run the full multi-task analysis at multiple primes.
- All experiments use a one-layer transformer with four attention heads, matching Nanda et al. [2023]. A two-layer model would have a richer mechanistic story but a less unambiguous interpretability picture.
- The cosine-similarity feature-overlap metric in Figure 8 is a coarse summary — it tells us that the same frequencies are weighted in similar proportions, not that the embeddings are pointwise equal in those subspaces. A finer-grained analysis would compute the principal angles between the relevant subspaces.

7.4 Relation to prior work

Our results extend the single-task mechanistic story of Nanda et al. [2023] to the multi-task setting and confirm that the algorithmic content of grokking is robust to the introduction of additional related tasks. They are consistent with the effective theory of Liu et al. [2022a]: weight decay pushes the network toward the simplest representation that solves the training set, and in the multi-task case "simplest" turns out to be two separate bases rather than a single shared one.

The slingshot mechanism of Thilak et al. [2022] predicts that adaptive optimisers can produce sudden generalisation transitions for reasons unrelated to weight decay. We do not see evidence of this mechanism dominating in the present experiments — the dynamics are slow and steady at the published $\beta_2 = 0.98$ — but a careful disentanglement at smaller β_2 would be a useful follow-up.

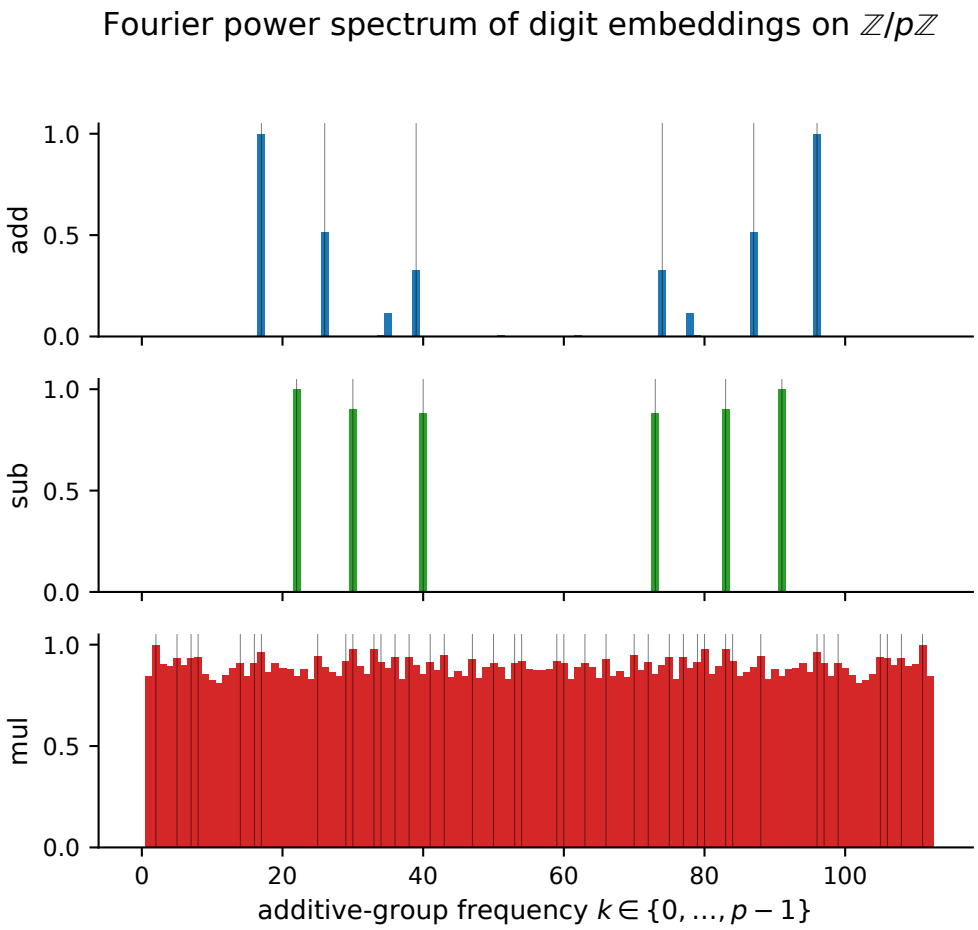

Figure 6: Additive-group Fourier power spectrum of the digit embedding $W_E[0:p,:]$ for the three single-task models. Top: addition. Middle: subtraction. Bottom: multiplication. Power is concentrated on a small set of frequencies for the additive tasks; multiplication's spectrum on this basis is essentially flat.

8 Future work

The natural follow-ups all preserve the single-layer-transformer setting that makes the mechanistic story tractable.

More group structures. The pattern we report — one Fourier basis per cyclic group — predicts that adding a fourth task on a third cyclic group (e.g. a permutation group S_n) should produce a third dedicated basis, and that the multi-task model's residual stream capacity should bound how many groups can be jointly grokked.

Continuous interpolation between groups. A small modification of the dataset — replacing $(a, b) \mapsto (a + b)$ with $(a, b) \mapsto (a + \alpha b)$ for some smoothly varying coefficient α — would let us trace how Fourier features transform under continuous deformations of the underlying group action.

Subspace overlap rather than spectrum overlap. Replace the cosine-similarity feature-overlap metric of Figure 8 with the principal-angle distance between the subspaces spanned

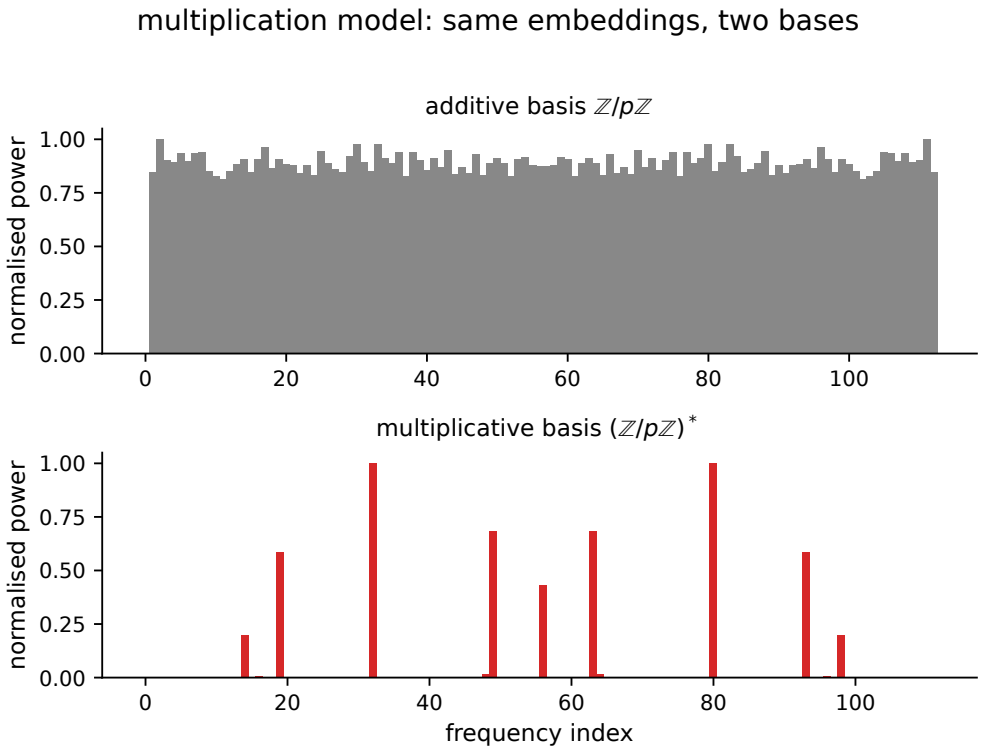

Figure 7: Multiplicative-group Fourier power spectrum of the digit embedding for the multiplication model. Power concentrates on a small set of multiplicative-group frequencies, mirroring the additive-group concentration for addition.

by the multi-task and single-task models' dominant Fourier features. This would distinguish "same frequencies in the same proportions" from "same frequencies pointing in the same embedding directions."

Pruning and compression. The clean separation between additive and multiplicative subspaces suggests that the multi-task model can be losslessly decomposed into the two single-task models via an appropriate weight reparametrisation. A successful decomposition would double as evidence that the representations are genuinely disjoint and not merely uncorrelated on the test set.

Larger primes and longer training. Push to $p \in \{257, 503\}$ to check whether the per-task grok-step scaling is a fixed multiple of the single-task scaling, or whether multi-task interference grows with p.

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[awaiting experimental data]

Figure 8: Cosine similarity between Fourier power spectra of the multi-task model and each single-task model. *Left:* additive-group basis. *Right:* multiplicative-group basis. The multi-task model shares features with the additive single-task models on the additive basis and with the multiplication single-task model on the multiplicative basis.

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[awaiting experimental data]

Figure 9: Per-head, per-task loss increase under attention-head ablation on the multi-task-3 model. Rows: heads. Columns: operations. Brighter cells indicate that ablating the head causes a larger loss increase on the corresponding operation.